

HW 3

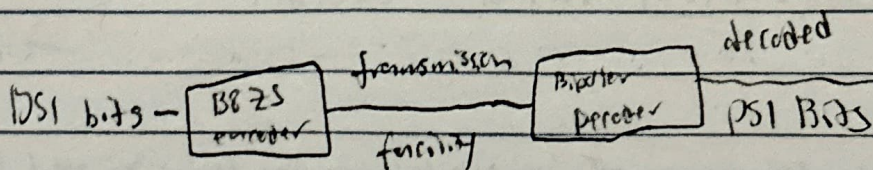
1. Determine the type of line coding implemented by this system. Z^{-1} box implements a delay of one bit.
for input b_n , $a_n = (b_n \oplus b_{n-1}) - b_{n-1}$

n	b_n	$b_n \oplus b_{n-1}$	b_{n-1}	a_n
0	0	0	0	0
1	1	1	0	1
2	1	0	1	-1
3	0	0	0	0
4	1	1	0	1
5	0	1	1	0
6	1	0	1	-1

This is a **bipolar-AMI**

b/c for every 1, polarity inverts.

2. B8ZS coded PS1 signal is connected to bipolar decoder:



* Refer to slide 7 wk 3 for extra context

PS1 input is rand seq w/ $\text{Pr}[0] = 1/4$, and transmission is error free. What is error rate in the decoded output? What is rate of bipolar violations (BPV) seen by receiver?

Note: on any a rand bit stream w/ $\text{Pr}[0] = p$, a seq of 8 consecutive zeros occurs once every m -bits

$$m = \frac{1-p^8}{1-p^8}$$

2 violations encoded when see 8-zeros,

00000000

B8ZS: 000V B0VB $\leftarrow$ 4 bits don't match

$$m = \frac{1 - .25^8}{1 - .25^9} = 87380 \text{ bits}$$

$$\frac{4 \text{ incorrect bits}}{87380 \text{ total}} = .00458\% \text{ error rate / bit}$$

$$\frac{2 \text{ BPV} \times 10^5}{87380} = .0022990 \text{ BPV rate / bit}$$

}, sine wave is used for PSK and QPSK, Duration = 10^{-5} s.

$$s(t) = .005 \sin(2\pi 10^6 t + \theta) V$$

measured noise power at receiver is 2.5×10^{-8} W.

determine E_b/N_0 (in dB) for PSK & QPSK.

For N_0 , we can assume Rx BW is reciprocal of duration
in this case $Rx BW = 10^5$ Hz

$$\text{PSK: } \frac{E_b}{N_0} \geq \frac{\text{SNR} \times \text{BW}}{\text{data rate}} \quad S = \frac{.005^2}{2} = 1.25 \times 10^{-5} \text{ W}$$

$$\text{SNR} = 1.25 \times 10^{-5} / 2.5 \times 10^{-8} = 500$$

$$\frac{E_b}{N_0} = \frac{SNR \cdot BW}{\text{Data rate}} \quad SNR = 500, BW = 10^5$$

$$\text{Data rate} = R = \frac{1}{T_s} = \frac{1}{10^{-5}} = 10^5$$

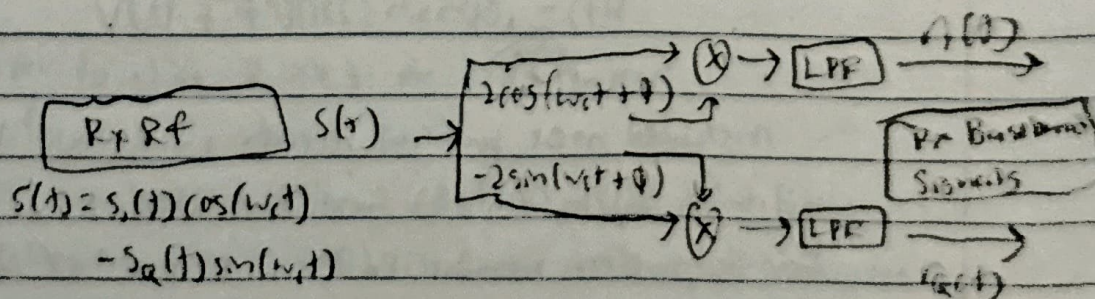
$$\frac{500 \times 10^5}{10^5} = 500 = \frac{E_b}{N_0} = 26.990 \text{ dB for PSK}$$

$$QPSK = \frac{SNR \cdot BW}{2R} \quad 2 \text{ bits per bit}_s, \text{ so } \frac{2}{T_s} = 200k = R$$

$$\frac{500 \times 10^5}{2(10^5)} = 250 = \frac{E_b}{N_0} = 23.979 \text{ dB for QPSK}$$

4.

QAM Demodulator. Rx carrier has phase offset ϕ compared to the transmit carrier. Substitution $\omega_c = 2\pi f_c$ to simplify notation



or, show that Rx in phase baseband output $r_I(t)$ is a combo of the Tx baseband signals $s_I(t), s_Q(t)$. derive the other $r_Q(t)$

$$2 \cos A \cos B = \cos(A+B) + \cos(A-B)$$

$$2 \cos A \sin B = \sin(A+B) - \sin(A-B)$$

$$r_1(t) = \text{LPF} \{ 2s(t) \cos(\omega_c t + \phi) \}$$

$$= \text{LPF} \{ (s_1(t) \cos \omega_c t - s_2(t) \sin \omega_c t) * 2 \cos(\omega_c t + \phi) \}$$

$$= \text{LPF} \{ 2 \cos(\omega_c t + \phi) s_1(t) \cos \omega_c t - 2 \cos(\omega_c t + \phi) s_2(t) \sin \omega_c t \}$$

$$= \text{LPF} \{ s_1(t) (\cos(2\omega_c t + \phi) + \cos \phi) - s_2(t) (\sin(2\omega_c t + \phi) - \sin \phi) \}$$

↑
killed by LPF

$$r_1(t) = s_1(t) \cos \phi + s_2(t) \sin \phi$$

b. For $\phi = \pi/4$, $r_1(t) = \frac{\sqrt{2}}{2} (s_1(t) + s_2(t))$

Best case: $\phi = 0$, s_1 fully through, s_2 killed

W/ $\phi = \pi/4$, our signal is attenuated by $\sqrt{2}/2$,
and the cross-talk is equally amplified

5. spectral components in range 300-3000 Hz

Sampling rate 2000 samples per sec used to generate
PCM signal

a. For $\text{SNR} \geq 30 \text{ dB}$, what # of uniform quant levels needed?

b. data rate required

$n = \text{bits}$

$$\text{SNR} = 6.02n + 1.76 \text{ dB} \rightarrow 30 \geq 6.02n + 1.76$$

$$n \geq 4.69 \rightarrow \text{round to } 5$$

$$7000 \text{ samples/s} \times 5 \text{ bits} = 35 \text{ kbps} \leftarrow \text{data rate (b)}$$

$$2^5 = 32 \text{ quantization levels (a)}$$